

**GRAPH THEORY:
HOMEWORK 6
BOOK SECTION 3.1 PART II.**

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1. Prove or disprove: Every tree has at most one perfect matching. (Hint: If a tree had two different perfect matchings, what could you say about the edges that belong to exactly one of them?)

Proof. The statement is true. Suppose, for a contradiction, that a tree T has distinct perfect matchings M_1 and M_2 . Consider the spanning subgraph with edge set

$$M_1 \triangle M_2 = (M_1 \setminus M_2) \cup (M_2 \setminus M_1).$$

At each vertex, the two perfect matchings either use the same edge, giving degree zero in this subgraph, or use different edges, giving degree two. Since the matchings are distinct, this subgraph has a component containing an edge. Every vertex of that component has degree two, so the component is a cycle. This contradicts the fact that a tree contains no cycle. Thus every tree has at most one perfect matching. $\square$

2. Prove that every maximal matching in a graph G has at least $\alpha'(G)/2$ edges. (Hint: Compare a maximal matching M with a maximum matching M^* . What does maximality of M tell you about each edge of the maximum matching?)

Proof. Let M be a maximal matching and let M^* be a maximum matching in G . Every edge of M^* has at least one endpoint saturated by M ; otherwise, that edge could be added to M , contradicting maximality.

For each edge of M^* , choose an edge of M sharing an endpoint with it. Each edge of M can be chosen by at most two edges of M^* , since it has two endpoints and at most one edge of the matching M^* is incident with each endpoint. Consequently,

$$\alpha'(G) = |M^*| \leq 2|M|,$$

and hence $|M| \geq \alpha'(G)/2$. $\square$

3. Prove that if G is an X, Y -bigraph such that

$$|N(S)| > |S| \quad \text{whenever } \emptyset \neq S \subsetneq X,$$

then every edge of G belongs to some matching that saturates X . (Hint: Given an edge xy , consider the graph $G' = G - x - y$. What would you like to show about G' , and how does deleting y affect neighborhoods?)

Proof. Fix any edge $xy \in E(G)$, where $x \in X$ and $y \in Y$. Let $G' = G - x - y$, with partite sets $X' = X \setminus \{x\}$ and $Y' = Y \setminus \{y\}$. We verify Hall's condition for X' . If $S \subseteq X'$ is nonempty, then S is a proper subset of X , so the hypothesis and integrality give $|N_G(S)| \geq |S| + 1$.

Deleting x does not affect the neighbors of S , and deleting y removes at most one neighbor. Therefore,

$$|N_{G'}(S)| = |N_G(S) \setminus \{y\}| \geq |N_G(S)| - 1 \geq |S|.$$

The inequality also holds for $S = \emptyset$. By Hall's Theorem, G' has a matching M' saturating X' . Since neither x nor y belongs to G' , the set $M' \cup \{xy\}$ is a matching in G that saturates X and contains xy . As xy was arbitrary, the conclusion follows. If G has no edges, the assertion about every edge is vacuous. $\square$

4. Use the König–Egerváry Theorem to prove that every bipartite graph G has a matching of size at least $e(G)/\Delta(G)$. Use this to conclude that every subgraph of $K_{n,n}$ with more than $(k-1)n$ edges has a matching of size at least k . (Hint: König–Egerváry lets you bound a different parameter instead of $\alpha'(G)$. How many edges can a single vertex take care of?)

Proof. Assume $\Delta(G) > 0$, as required for the stated ratio to be defined. Let Q be a minimum vertex cover of G . By the König–Egerváry Theorem, $|Q| = \alpha'(G)$. Since every edge has at least one endpoint in Q ,

$$e(G) \leq \sum_{v \in Q} d_G(v) \leq |Q|\Delta(G) = \alpha'(G)\Delta(G).$$

Dividing by $\Delta(G)$ yields $\alpha'(G) \geq e(G)/\Delta(G)$. For an edgeless graph, the empty matching has size zero; the division-free inequality above still holds.

Now let G be a subgraph of $K_{n,n}$ with $e(G) > (k-1)n$, where n and k are positive integers. Then G has at least one edge and $\Delta(G) \leq n$, so

$$\alpha'(G) \geq \frac{e(G)}{\Delta(G)} \geq \frac{e(G)}{n} > k-1.$$

Since $\alpha'(G)$ is an integer, $\alpha'(G) \geq k$. $\square$

5. Let G be an X, Y -bigraph with vertex cover Q , and let H be the induced $(X-Q), (Q \cap Y)$ -sub-bigraph of G . Prove that if no matching in H saturates $Q \cap Y = Q - X$, then G has a vertex cover that is smaller than Q . (Hint: What does Hall's Theorem tell you when no matching in H saturates $Q - X$? Use that to decide which vertices of Q you might remove and replace to obtain a smaller set, and explain why the new set is a vertex cover.)

Proof. Since no matching in H saturates $Q \cap Y$, Hall's Theorem gives a set $S \subseteq Q \cap Y$ such that

$$|N_H(S)| < |S|.$$

Define

$$Q' = (Q \setminus S) \cup N_H(S).$$

Because $N_H(S) \subseteq X \setminus Q$, the two sets in this union are disjoint, and

$$|Q'| = |Q| - |S| + |N_H(S)| < |Q|.$$

It remains to show that Q' is a vertex cover of G . Let uv be any edge, with $u \in X$ and $v \in Y$. If $v \notin S$, at least one endpoint lies in Q , and neither endpoint lies in S ; thus an endpoint remains in $Q \setminus S$. If $v \in S$ and $u \in Q$, then $u \in Q \setminus S$, since $S \subseteq Y$. Finally, if $v \in S$ and $u \notin Q$, then uv is an edge of H , so $u \in N_H(S)$. In every case, uv has an endpoint in Q' . Hence Q' is a vertex cover smaller than Q , as desired. $\square$